

Mathematics of Data Science - CA 2

Name: _____

A tech company fabricates qubits. Each qubit’s reliability is tested and the outcome is modelled with a Bernoulli random variable X . The outcome is either **Good** ($X = 1$) or **Faulty** ($X = 0$). The fabrication process has two states: **Stable (A)** and **Unstable (B)**.

- The process is **Stable** with probability $\mathbb{P}(A) = 2/3$.
 - When Stable, the probability of a qubit being **Good** is $\mathbb{P}(X = 1|A) = 0.9$.
 - When Unstable, the probability of a qubit being **Good** is $\mathbb{P}(X = 1|B) = 0.6$.
1. (a) Prove that the overall probability that a randomly produced qubit is Good is $p = 0.8$.
(b) You test one qubit and find that it is Good. What is the updated probability that the fabrication process was in the **Stable** state?

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4 pts

2. You are a financial analyst at the company. You assign a monetary value to each qubit produced:
- A **Good** qubit generates a profit of **4 CHF**.
 - A **Faulty** qubit results in a loss of **1 CHF** (a value of -1 CHF).

Let V be the random variable representing the financial value of a single qubit. What is the expected financial value $\mathbb{E}[V]$, of a randomly chosen qubit?

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3 pts

3. You take a sample of $n = 100$ qubits. Let S_{100} be the total number of Good qubits in the sample. Use the **Central Limit Theorem (CLT)** to estimate the probability that you find **84 or more Good** qubits. Express this probability in terms of the Cumulative Distribution Function (CDF) of the standard normal distribution, $\Phi(z)$.

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3 pts



1. (a) **Overall Probability (p):**

Using the Law of Total Probability:

$$p = \mathbb{P}(X = 1) = \mathbb{P}(X = 1|A)\mathbb{P}(A) + \mathbb{P}(X = 1|B)\mathbb{P}(B) = 0.9 \cdot \frac{2}{3} + 0.6 \cdot \frac{1}{3} = \frac{1.8}{3} + \frac{0.6}{3} = \frac{2.4}{3} = 0.8.$$

(b) **Updated Probability (Bayes' Theorem):**

We want to find $\mathbb{P}(A|X = 1)$. Using Bayes' Theorem:

$$\mathbb{P}(A|X = 1) = \frac{\mathbb{P}(X = 1|A)\mathbb{P}(A)}{\mathbb{P}(X = 1)} = \frac{0.9 \cdot 2/3}{4/5} = \frac{9}{12} = 0.75.$$

2. We use the law of total expectation, conditioning on the state of the fabrication process (A: Stable or B: Unstable).

$$\mathbb{E}[V] = \mathbb{E}[V|A] \cdot \mathbb{P}(A) + \mathbb{E}[V|B] \cdot \mathbb{P}(B)$$

(a) **Calculate the conditional expectation in the Stable state (A):**

When the process is Stable, the probability of a qubit being Good is $\mathbb{P}(X = 1|A) = 0.9$, and Faulty is $\mathbb{P}(X = 0|A) = 1 - 0.9 = 0.1$.

$$\mathbb{E}[V|A] = 4 \cdot \mathbb{P}(X = 1|A) - 1 \cdot \mathbb{P}(X = 0|A) = 4 \cdot 0.9 - 1 \cdot 0.1 = 3.6 - 0.1 = 3.5$$

(b) **Calculate the conditional expectation in the Unstable state (B):**

When the process is Unstable, the probability of a qubit being Good is $\mathbb{P}(X = 1|B) = 0.6$, and Faulty is $\mathbb{P}(X = 0|B) = 1 - 0.6 = 0.4$.

$$\mathbb{E}[V|B] = 4 \cdot \mathbb{P}(X = 1|B) - 1 \cdot \mathbb{P}(X = 0|B) = 4 \cdot 0.6 - 1 \cdot 0.4 = 2.4 - 0.4 = 2$$

(c) **Combine using the law of total expectation:**

$$\mathbb{E}[V] = 3.5 \cdot 2/3 + 2 \cdot 1/3 = 7/3 + 2/3 = 9/3 = 3\text{CHF}$$

3. Let X_i be the Bernoulli random variable for the i -th qubit, where $X_i = 1$ if it is Good. From question 1(a), the probability of a single qubit being Good is $p = \mathbb{P}(X_i = 1) = 0.8$. The total number of Good qubits is $S_n = \sum_{i=1}^n X_i$ with $n = 100$.

The Central Limit Theorem states that for large n , the sum S_n is approximately normally distributed.

The mean of the sum is $\mu = \mathbb{E}[S_n] = n \cdot p$.

$$\mu = 100 \cdot 0.8 = 80$$

The variance of the sum is $\sigma^2 = \text{Var}(S_n) = n \cdot p(1 - p)$.

$$\sigma^2 = 100 \cdot 0.8 \cdot (1 - 0.8) = 100 \cdot 0.8 \cdot 0.2 = 16.$$

The standard deviation is $\sigma = \sqrt{16} = 4$.

So, by the CLT, $S_{100} \approx N(\mu = 80, \sigma^2 = 16)$.

We want to find the probability $\mathbb{P}(S_{100} \geq 84)$. We convert this to the standard normal variable $Z = \frac{S_n - \mu}{\sigma}$.

$$Z = \frac{84 - 80}{4} = \frac{4}{4} = 1$$

The probability $\mathbb{P}(S_{100} \geq 84)$ becomes $\mathbb{P}(Z \geq 1)$.

$$\mathbb{P}(Z \geq 1) = 1 - \mathbb{P}(Z < 1) = 1 - \Phi(1).$$

The estimated probability of finding 84 or more Good qubits is $1 - \Phi(1)$.